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B. A Football Match

time limit per test: 2 seconds

memory limit per test: 1024 megabytes

There is an intense football match going on between Robots and Humans. However, things aren't as simple as they seem — the Robots have disguised themselves to look exactly like Humans! From the outside, it's impossible to tell who is a Robot and who is a Human.

The audience know only one important information — the Robots tackles only the Humans, and the Humans tackles only the Robots.

Now, you are given a list of tackles, each involving two players. Based on this information, find the maximum possible number of Robots or Humans.

Input

The first line contains two integers N and M ($1 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5$) — the number of players in the match and the number of tackles occurred during the match respectively.

The next M lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — player u_i tackled player v_i . Each tackle between two players will be reported at most once.

Output

Print the maximum possible number of Robots or Humans.

Examples

input

Copy

```
5 6
3 4
3 2
5 4
5 2
4 1
1 2
```

output

Copy

```
3
```

input

Copy

```
5 4
4 3
1 3
3 2
3 5
```

output

Copy

```
4
```

input

Copy

```
4 1
1 3
```

CSE221 Assignment 06 Summer 2026

Contest is running

3 days

Contestant


[→ Submit?](#)

 Language: Java 21 64bit

 Choose file: Choose File No file chosen

Success!



Submit

output

Copy

3

input

Copy

```
6 6
1 3
1 4
3 6
4 6
4 5
6 2
```

output

Copy

3

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The only programming contests Web 2.0 platform
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